

Reviewer Pack — Non-Commutative Rank Lower Bound for Disjointness Matrices

Entry f46ef08c667f · INCONCLUSIVE

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Non-Commutative Rank Lower Bound for Disjointness Matrices

Entry ID: f46ef08c667f **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-19 06:20:36 UTC

1. Conjecture

Field A (mathematical branch): COMMUNICATION_COMPLEXITY **Field B** (complexity object): NONCOMMUTATIVE_RANK

Statement:

The non-commutative rank of the disjointness matrix M_n (where $M_n[x][y] = 1$ iff $x \cap y = \emptyset$) satisfies $\text{rank}_{\{nc\}}(M_n) = \Omega(n)$.

Rationale (proposer's reasoning):

Non-commutative rank connects to algebraic branching programs (ABPs) via Barrington's theorem, which equates NC^1 circuits to width-5 ABPs. The disjointness matrix's high non-commutative rank would imply exponential ABP size, translating to $\Omega(n)$ communication complexity via the log-rank conjecture. This avoids algebraicization barriers by focusing on matrix structure rather than polynomial identities.

Taxonomy category: BARRINGTON_ALG (status at proposal time: partially_alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 694a4d92e42a96db

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.95	SAFE	SAFE
NATURAL_PROOFS	SAFE	0.95	SAFE	SAFE

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def gaussian_elimination(A):
    n = len(A)
    for i in range(n):
        # Find pivot row
        max_row = i
        for k in range(i+1, n):
            if abs(A[k][i]) > abs(A[max_row][i]):
                max_row = k
        A[i], A[max_row] = A[max_row], A[i]

        # Eliminate non-pivot elements
        pivot = A[i][i]
        for j in range(n):
            A[i][j] /= pivot

        for k in range(n):
            if k != i:
                factor = A[k][i]
                for j in range(n):
                    A[k][j] -= factor * A[i][j]

def non_commutative_rank(A):
    n = len(A)
    rank = 0
    for i in range(n):
        if all(abs(A[j][i]) < 1e-9 for j in range(n)):
            continue
        rank += 1
    return rank

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = random.randint(5, 40)
    M = [[Fraction(1 if i != j else 0, 1) for j in range(n)] for i in range(n)]
```

```

rank_nc = non_commutative_rank(gaussian_elimination(M))

return {
    "metric_name": "non-commutative rank",
    "metric_value": rank_nc,
    "instances_tested": n,
    "conjecture_holds": rank_nc >= n / 10, # Using a small constant c > 0
    "counterexample": "" if rank_nc >= n / 10 else f"Disjointness matrix M_{n} has non-commutative rank > n/10"
}

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    results = []

    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = math.sqrt(sum((result["metric_value"] - mean_value) ** 2 for result in results) / len(results))
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(not result["conjecture_holds"] for result in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"Disjointness matrix has non-commutative rank > n/10\"")
    else:
        print("RESULT: INCONCLUSIVE insufficient evidence")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b5ffe8ca.py", line 68, in <module>
    trial_result = run_trial(seed)
                   ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b5ffe8ca.py", line 53, in run_trial
    rank_nc = non_commutative_rank(gaussian_elimination(M))
               ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b5ffe8ca.py", line 40, in non_commutative_rank
    n = len(A)
        ~~~~
TypeError: object of type 'NoneType' has no len()

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed due to `TypeError`, preventing data collection. The pre-registered support condition cannot be evaluated without valid results. | next: Fix the `non_commutative_rank` function to handle `NoneType` inputs and re-run tests with multiple seeds

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	84357
2	preregistration	ollama_remote	qwen3:8b	0	0	27510
3	novelty	ollama_remote	qwen3:8b	0	0	24508
4	novelty	ollama_remote	qwen3:8b	0	0	16938
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10370
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8293
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8926
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8697
9	judge	ollama_remote	qwen3:8b	0	0	20344

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 209943 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/f46ef08c667f.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/f46ef08c667f.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/f46ef08c667f.tar.gz` (if generated)